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College of Informatics and Computing Sciences

IT 322: ADVANCED SYSTEMS INTEGRATION AND ARCHITECTURE

SEMESTRAL PROJECT: Enterprise-Grade System Development

I. PROJECT OVERVIEW

The Semestral Project for IT 322 is centered on the technical advancement of the **approved Capstone proposal**. The primary objective is the development and presentation of a functional system prototype operating as a **3-Tier Integrated System**. This requirement ensures the solution adheres to industry-standard interoperability, security, and scalability patterns.

Rigorous application of these architectural standards at this stage significantly reduces technical debt and identifies potential integration issues early. Completion of this project ensures the system is functionally sound for IT 322 and provides a robust, **defense-ready foundation for the final Capstone period**. Successfully implementing these enterprise standards facilitates a smoother transition into the actual Capstone process, minimizing revisions and technical hurdles during the final deliberation.

II. OUTCOMES AND GOALS

A. Objective To collaboratively develop and present a functional system prototype demonstrating advanced systems integration and architectural separation. This process serves as a critical technical stress test for the Capstone, ensuring that core features are supported by an enterprise-ready infrastructure.

B. Intended Learning Outcomes (ILOs) By the end of the project, a software prototype will be developed that:

- **ILO 1:** Utilizes architectural design tools to implement a strictly separated 3-Tier Architecture (Presentation, Business, and Data).
- **ILO 2:** Implements advanced integration strategies using Web APIs, Hardware components, or external data sources to ensure seamless system interoperability.
- **ILO 3:** Applies professional security and data integrity standards to ensure system resilience in high-stakes defense environments.

III. MAJOR MILESTONES & DELIVERABLES

A. The Consolidated Technical Portfolio Groups must submit a single consolidated document to serve as the official technical reference for evaluators. As the three (3) required Materials have already been completed during the semester, this task involves consolidating those files and providing necessary updates to reflect the latest progress of the prototype.

- **Deadline:** The updated consolidated document must be submitted at least **two (2) days prior** to the scheduled presentation day.
- **Strategic Benefit:** This portfolio establishes the technical baseline for Capstone documentation, ensuring architectural decisions are vetted and documented prior to the final Capstone defense.



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B. Final Prototype Defense and System Cycle To be considered a valid prototype, the demonstration must complete an **entire cycle of processing** within the system. The primary objective is to complete a full test of the project flow without interruptions.

1. **System Cycle:** The demonstration must show data traveling from initial input (Tier 1), through business logic processing (Tier 2), to final storage or output (Tier 3).
2. **Use of Mockups:** If specific external integrations, such as hardware components or third-party APIs, are not yet fully operational, the use of **mockups or temporary data** is permitted. However, the logic path and intended behavior for these integrations must be clearly demonstrated.
3. **Core Proofs:**
 - **Interface Interaction:** Initiation of core functions from Tier 1.
 - **Logic Processing:** Verification of Tier 2 management of business rules and data validation.
 - **Integration Flow:** Successful execution of a live handshake or a clearly defined mocked integration.
 - **Secure Finalization:** Verification of validated and secure data storage in Tier 3.

IV. TECHNICAL EVALUATION REQUIREMENTS

Evaluators will cross-reference the live demonstration with the Technical Portfolio. Mastery in the following areas is required to prevent common technical failures encountered during Capstone defenses:

- **Architectural Integrity:** Explicit proof that the Presentation Tier does not communicate directly with the Data Tier (Database).
- **Data Integrity:** Demonstration of the ability to reject invalid data at the logic layer before it is committed to storage.
- **Security Implementation:** Verification that sensitive data (e.g., passwords) is hashed and that API keys or credentials are kept in secure configuration files rather than being hard-coded into the logic.

General Reminders:

- **Technical Readiness:** Multiple devices must be prepared with the complete environment setup (IDE, Database, Git Repository) to prevent technical delays.
- **Capstone Advantage:** Proving the system's architectural soundness now ensures it is more "defense-ready" for the Capstone period, significantly reducing the risk of critical failures during the final requirements check.